The surveillance of tick bites on humans and additional research opportunities through citizen science in Belgium

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Ticks are known to carry a wide variety of tick-borne pathogens, posing a potential health risk to humans. In 2015, a citizen science surveillance platform, TekenNet/TiquesNet, was launched in order to better understand the temporal and spatial risk of tick bites on humans in Belgium (A). The platform is also used for additional research (B, C and D).

Methods: TekenNet/TiquesNet - Citizen Surveillance

- Reports of tick bites on humans in Belgium through a website and mobile application
- Information collected includes date and context of the bite(s) (type of environment, location, activity)

A. Epidemiology of human tick bites

Context of tick bite notifications (mean 2016-2024):

- **Environment**: 44.2% in gardens, 36.5% in forest
- **Distance to home**: 77.6% within 10 km
- **Activity**: 88.0% during leisure activities

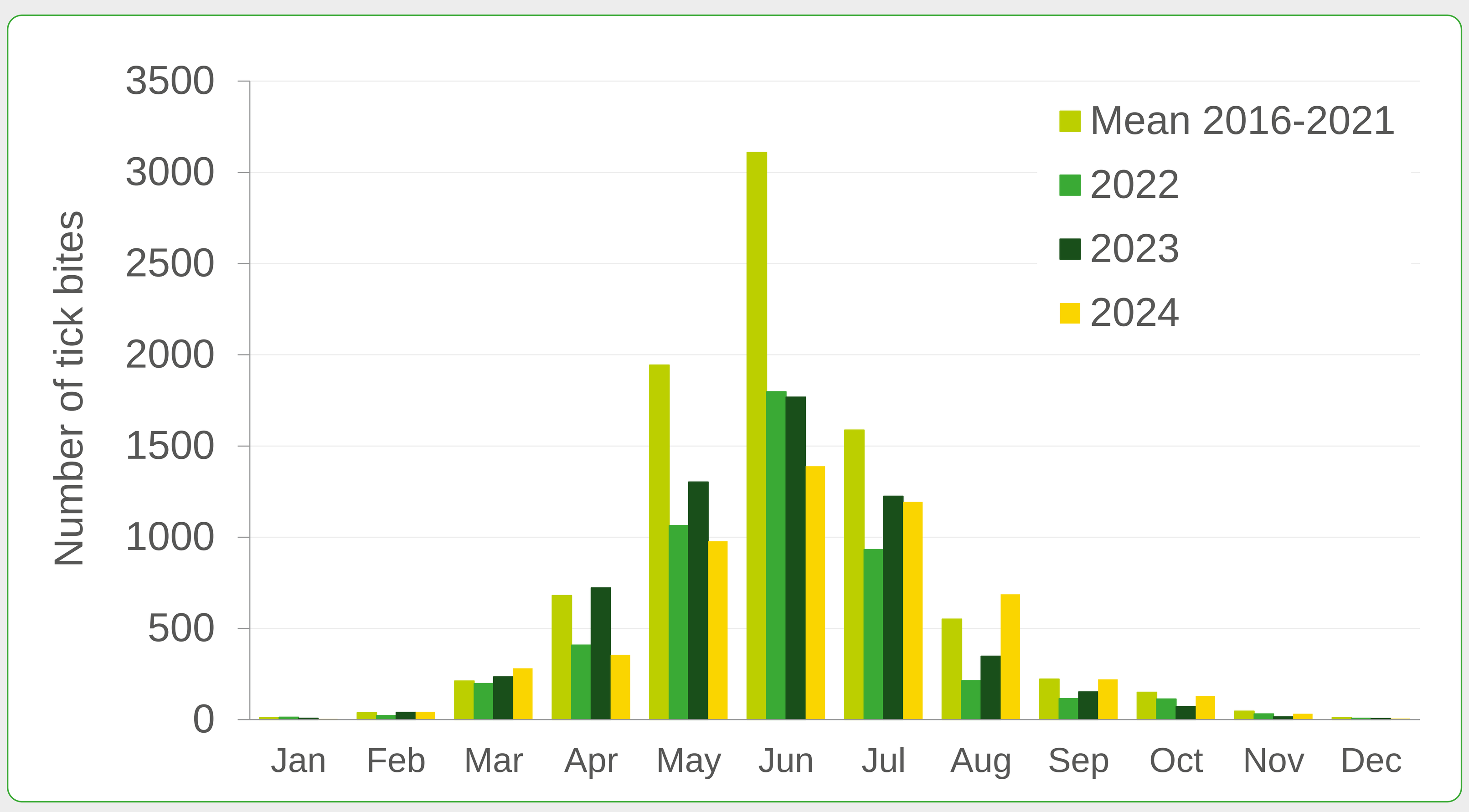
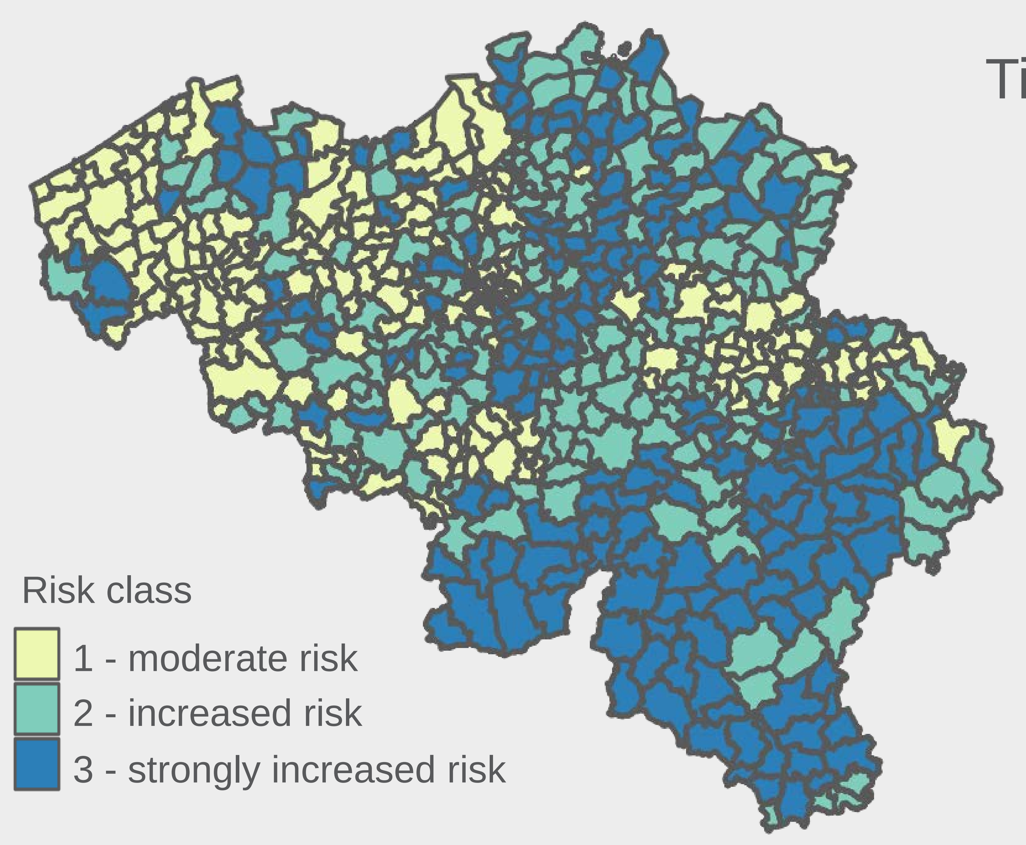


Figure 1: Number of reported tick bites per month and per year

B. Risk map

- Bayesian modelling of **tick bite data**
- Includes **environmental predictors**

⇒ Support implementation of **preventive measures** for municipalities



Tick bite associated variables

- Deciduous forest
- Coniferous forest
- Average density forest edges
- Vegetation index (NDVI)
- Deer presence
- Recreational zones

Figure 2: Risk map for Belgian municipalities based on a spatial Bayesian hierarchical model. Classes are defined as the probability that the risk of attracting tick bites in a certain municipality is higher than what is expected average in Belgium.

C. Prevalence of tick-borne pathogens in ticks

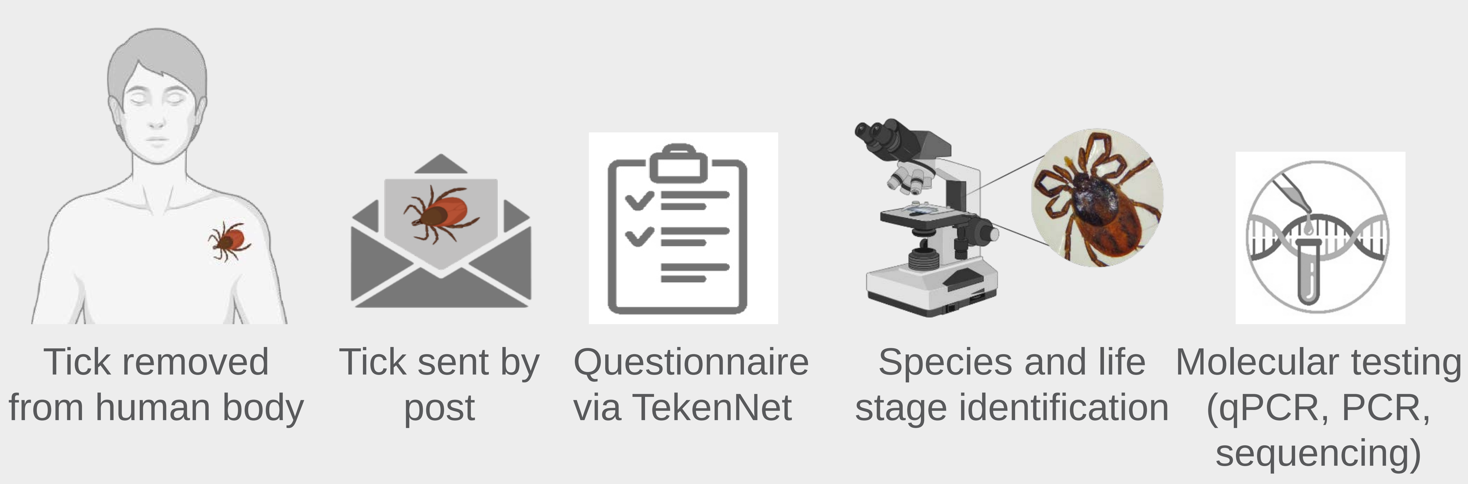


Figure 3: Method tick collection and laboratory analyses

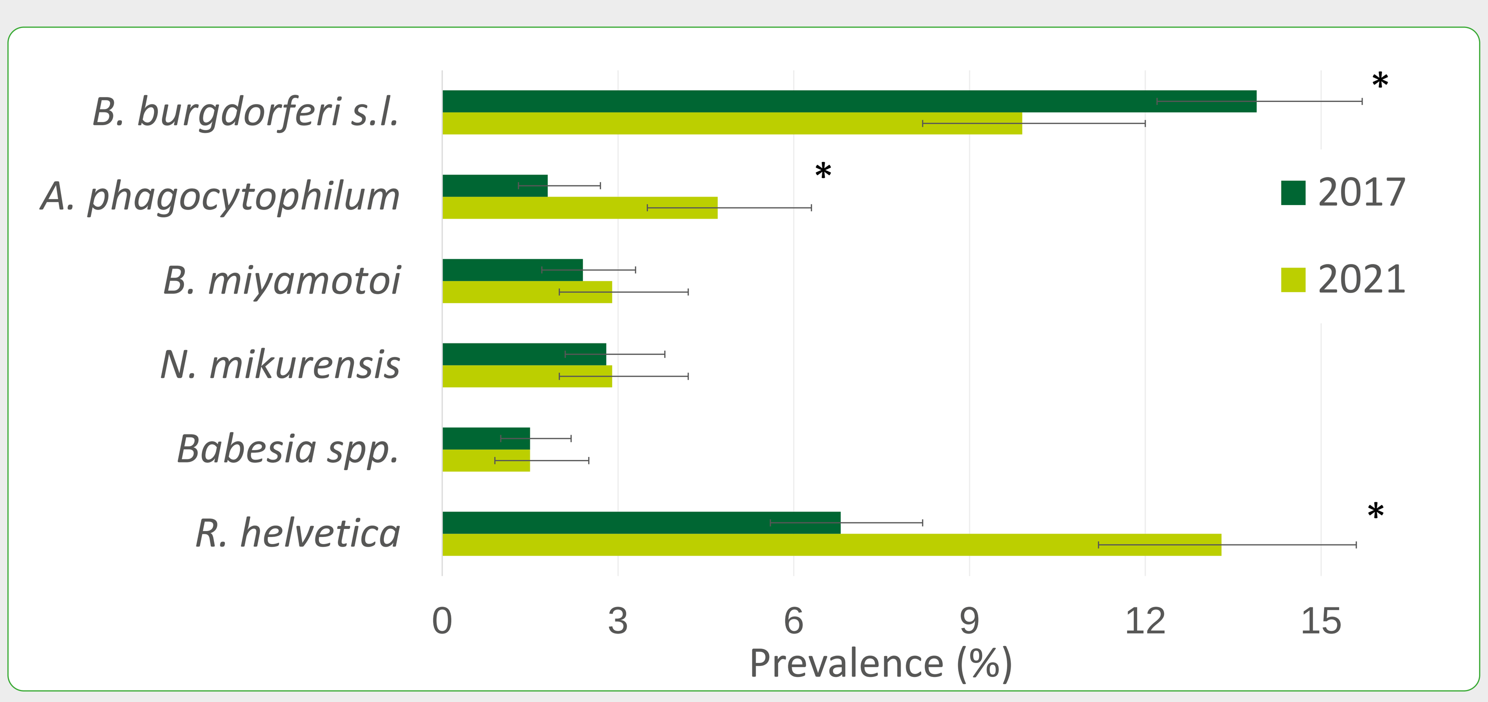


Figure 4: Presence of pathogens in ticks collected from humans in 2017 (n = 1,515) and 2021 (n = 928). An * indicates a statistical difference (p-value < 0.05)

D. Collection of ticks in gardens (2025-2028)

- Research **density** of (infected) ticks in private gardens
- Impact **weather conditions** and **vegetation types**

⇒ Guide prevention policies
⇒ Anticipate on effect of future climate change

Citizens will be invited to:

- Build a tick flag
- Collect ticks min. 3x/year (1x/month) during the tick season
- Fill in a questionnaire
- Send ticks to Sciensano → identification and tick-borne pathogen analysis

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